ARJUN SINGH

Systems-Oriented Software Engineer & Technical Product Builder

Jaipur, Rajasthan, India • arjunsingh.dev@outlook.com • linkedin.com/in/arjun-singh-682453378 • github.com/wdymarjun • unstop.com/u/arjunsin14938

PROFESSIONAL SUMMARY

High-velocity 1st-year Computer Science engineer and 3-time Hackathon Team Lead with deep technical expertise in advanced Lua architecture, algorithmic C++, and AI prompt engineering. Combines rigorous mathematical intuition with an end-to-end product builder mindset, engineering resilient low-latency systems and translating complex backend architectures into scalable, high-impact user experiences.

EDUCATION

Rajasthan Institute of Engineering and Technology (RIET) | B.Tech in Computer Science & Engineering — Expected 2030 (1st Year) | Jaipur, Rajasthan

Relevant Foundations: *Engineering Mathematics, Discrete Mathematics, Algorithmic Complexity (O(n) Analysis), Object-Oriented Programming (C++), Spatial Geometry & Matrix Transformations*

TECHNICAL & SYSTEM SKILLS

Languages & Core:	**Advanced Lua (Luau)** (Metatables, OOP, Server-Client Replication, Network Optimization), **C++** (OOP, Pointers, Manual Memory Management, STL, Data Structures & Algorithms).
AI & Prompt Engineering:	Prompt Architecture, Context Window Optimization, LLM Workflow Automation, Chain-of-Thought System Design, Model Evaluation & Benchmarking.
Engine & Systems:	Roblox Studio Engine (Advanced 3D Spatial UX, Client-Server Synchronization, Physics Optimization, Memory Profiling, Anti-Cheat & Exploit Mitigation Logic).
Mathematical Foundations:	Engineering Mathematics, Discrete Structures, Algorithmic Complexity ($O(n)$ analysis), Spatial Geometry, Matrix Transformations.
Tools & Workflows:	Git, GitHub (Branch Strategy, Version Control), Linux/CLI, VS Code, Agile Sprint Planning, Rapid Prototyping.

COMPETITIVE HACKATHONS & LEADERSHIP

3x Technical Team Lead | Competitive Hackathon Circuit — 2026

- Spearheaded multi-disciplinary engineering squads across 3 rapid hackathons, owning sprint backlogs, task delegation, technical architecture, and live pitches delivered to senior industry evaluation panels.

Smart India Hackathon (SIH) – Internal College Round | Ranked 5th of 40+ Teams — RIET Jaipur | 2026

- Formulated and pitched a high-viability technical architecture under strict 24-hour constraints; achieved 5th position overall out of 40+ competing engineering teams for system feasibility and technical rigor.

HackPulse Hackathon | Secured 4th Position Overall — UEM Jaipur | 2026

- Engineered a functional production prototype within 36 hours; defended decoupled client-server architecture and data flow before enterprise judges, securing 4th place across regional institutes.

Pitchathon by Kalshinax Company | Awarded 2nd Runner-Up — RIET Jaipur | 2026

- Authored technical validation and go-to-market architecture for an innovative software solution; placed 2nd Runner-Up based on product viability metrics, unit economics, and live execution.

FEATURED TECHNICAL PROJECTS

Advanced 3D Spatial Engine & Game Architecture | Roblox Studio, Luau, Metatables — Interactive 3D Engine

- Architected an atmospheric, high-fidelity 3D immersive world (ASC) utilizing custom modular assets, tailored character animation pipelines, and dynamically tuned atmospheric lighting systems.
- Engineered robust server-client replication using Luau metatables and RemoteEvents/RemoteFunctions, guaranteeing sub-45ms network sync and strict server-side state validation against unauthorized manipulation.
- Optimized runtime memory and client frame rates, reducing memory overhead by 38% and eliminating frame stutter through custom part-caching, collision-matrix tuning, and spatial streaming.

Algorithmic Systems & Data Structures Implementation | Modern C++, STL — Systems Engineering

- Implemented core data structures and algorithmic patterns (self-balancing trees, dynamic arrays, graph traversal, custom allocators) from scratch in C++ with explicit pointer arithmetic and memory profiling.
- Benchmarked computational trade-offs using discrete mathematics and Big-O asymptotic analysis, optimizing worst-case operations from $O(n^2)$ to $O(n \log n)$ while eliminating memory fragmentation.

Hackathon Production Build & AI Pipeline | Full-Stack Prototype, LLM Integration — Hackathon Release

- Directed architecture and integration sprints to ship an end-to-end prototype addressing real-world operational challenges under tight 36-hour hackathon deadlines.
- Integrated structured prompt engineering pipelines and automated LLM workflows with context window optimization, accelerating intake data synthesis by 65% with zero degradation in extraction accuracy.